

QTPTC1 Module 3

Anthony Slawski

November 2025

Qiskit Patterns

This lab used the Qiskit Patterns framework to build and analyze a Bell state. The Estimator primitive was used to study expectation values of a set of two qubit Pauli observables, labeled IZ , IX , ZI , XI , ZZ , and XX . For an ideal Bell state, only the joint observables ZZ and XX are expected to give nonzero expectation values (close to one), while the single qubit terms vanish. This pattern of correlations appeared in the results and matches the expected behavior of an entangled state, since each qubit gives outcomes with equal probability, while their joint outcomes are perfectly correlated.

The same logical circuit was then transpiled for different backends, producing distinct ISA level circuits that all implement the same Bell preparation. Using the Sampler primitive, measurement statistics were collected and showed counts concentrated on the correlated outcomes, up to small noise effects. Finally, the workflow was repeated on a real IBM Quantum backend, where the observed measurement distribution closely matched the simulator output.

Quantum simulation

This lab used Trotterization to simulate the real time dynamics of the transverse field Ising model on both simulators and real quantum hardware. The model was implemented as a sum of Pauli strings acting on linear chains of spins, with nearest neighbor ZZ couplings and a field of strength h at an angle α in the xz plane. The Hamiltonian was mapped to a sparse Pauli representation, and an initial configuration $|001100\rangle$ was prepared to study how spins evolve under the combined action of interactions and field.

Time evolution was approximated with Pauli evolution gates built from first order Lie-Trotter ($e^{-i(H_1+H_2)t} \approx e^{-iH_1t}e^{-iH_2t}$) and second ($S_2(t) = e^{-i(H_1+H_2)t} \approx e^{-iH_1t/2}e^{-iH_2t}e^{-iH_1t/2}$) and fourth ($S_4(t) = S_2(pt)^2S_2((1-4p)t)S_2(pt)^2$, with $p = \frac{1}{4-4^{1/3}}$) order Suzuki-Trotter product formulas, which decompose $e^{-i(H_1+H_2)t}$ into sequences of simpler exponentials. The magnetization and mean spin correlation were defined as averaged Z and nearest neighbor ZZ observables, and a statevector estimator was used to compute their expectation values at each time step (without collapsing the state).

For the six site chain, the energy remained nearly constant, as expected for a time independent Hamiltonian, while magnetization and spin correlations oscillated in time in a way set by the field strength and interactions. Comparing first and second order formulas to exact reference data showed that higher order Trotterization follows the exact curves more closely at the same time step, but at the cost of much deeper and more complex circuits.

The same approach was then applied to a two site lattice with an initial $|10\rangle$ state and different field angles. For a purely transverse field the dynamics mixed $|10\rangle$ and $|01\rangle$ and produced an almost complete swap at the chosen evolution time, while for a longitudinal field the state remained the same. Intermediate angles produced partial mixing with angle dependent swap probabilities. These patterns were first examined with an ideal statevector simulation and then reproduced on IBM Quantum hardware. The hardware results matched the simulated probabilities within noise, confirming that Trotterized time evolution accurately reflects field dependent dynamics of small spin systems. The fourth order decomposition was even more accurate in principle, but its large circuit depth makes it less useful for current hardware, where noise would dominate any gain in approximation order.